

Name: _____

Name and classify.

1. Name the polygon with 7 sides. With 10 sides. With 12 sides.
2. A stop sign is a regular octagon. What two things does the word REGULAR tell you about its sides and angles?
3. A polygon has a diagonal that passes outside the polygon. Is it convex or concave?

Interior angles.

4. Find the sum of the interior angle measures of a convex octagon.
5. Find the measure of each interior angle of a regular decagon.
6. The sum of the interior angle measures of a convex polygon is 1620° . How many sides does it have?
7. Four angles of a pentagon measure 92° , 118° , 107° , and 99° . Find the fifth angle.

Exterior angles.

8. Find the measure of each exterior angle of a regular nonagon.
9. The exterior angles of a quadrilateral measure $6t^\circ$, $9t^\circ$, $11t^\circ$, and $10t^\circ$. Find t and the largest exterior angle.
10. Each exterior angle of a regular polygon measures 24° . How many sides does the polygon have?